

Avat3r: Large Animatable Gaussian Reconstruction Model for High-fidelity 3D Head Avatars

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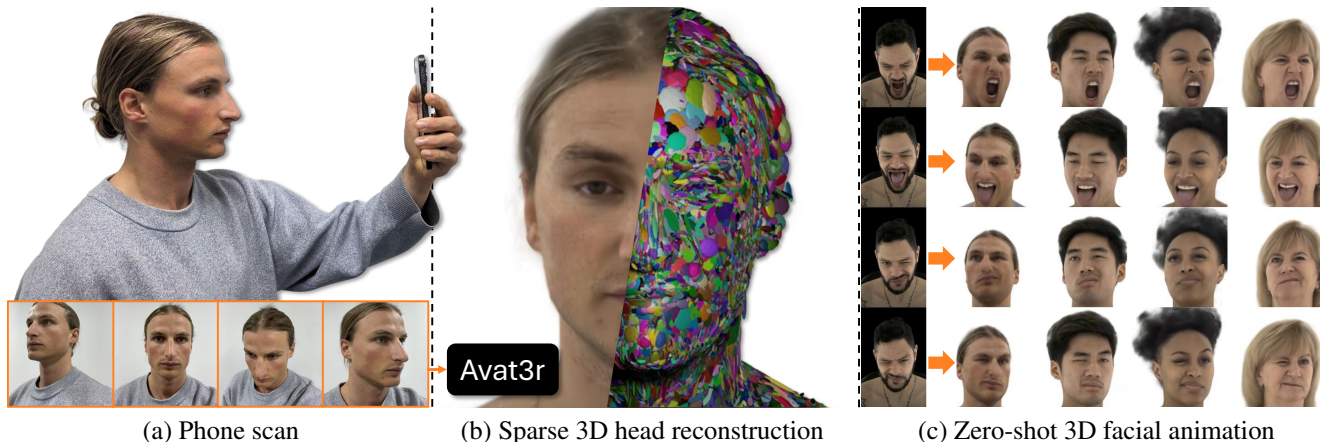


Figure 1. **Avat3r**. Given just four images of a person’s head (a), Avat3r achieves two things: (b) it creates a faithful 3D reconstruction of the head in a feed-forward manner, and (c) it allows facial animation without having seen any of the corresponding expressions of the person. This simplifies the capturing process as it drops the need for recording long sequences of facial movement. As a result, the entire pipeline from head scan to final 3D head avatar can be executed within a few minutes and runs on a single consumer-grade GPU.

Abstract

Traditionally, creating photo-realistic 3D head avatars requires a studio-level multi-view capture setup and expensive optimization during test-time, limiting the use of digital human doubles to the VFX industry or offline renderings. To address this shortcoming, we present Avat3r, which regresses a high-quality and animatable 3D head avatar from just a few input images, vastly reducing compute requirements during inference. More specifically, we make Large Reconstruction Models animatable and learn a powerful prior over 3D human heads from a large multi-view video dataset. For better 3D head reconstructions, we employ position maps from DUST3R and generalized feature maps from the human foundation model Sapiens. To animate the 3D head, our key discovery is that simple cross-attention to an expression code is already sufficient. Finally, we increase robustness by feeding input images with different expressions to our model during training, enabling the

reconstruction of 3D head avatars from inconsistent inputs, e.g., an imperfect phone capture with accidental movement, or frames from a monocular video. We compare Avat3r with current state-of-the-art methods for few-input and single-input scenarios, and find that our method has a competitive advantage in both tasks. Finally, we demonstrate the wide applicability of our proposed model, creating 3D head avatars from images of different sources, smartphone captures, single images, and even out-of-domain inputs like antique busts.

1. Introduction

Lifelike 3D head avatars find use in a variety of applications such as telepresence, movie production, personalized games, or digital education. Typically, the creation of a high-fidelity avatar requires studio-quality videos that have to be recorded in hour-long sessions in expensive capture setups. This limits the applicability of a 3D head avatar creation pipeline, especially in casual settings where users may only want to use a few images taken with their smartphone to create a convincing digital double. In this paper,

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Project website: <https://tobias-kirschstein.github.io/avat3r/>

we present Avat3r, a method that can reconstruct an animatable 3D head avatar from just a few images of a person.

Building such a system involves several hard challenges. First, sparse 3D reconstruction poses a heavily under-constrained optimization problem since some regions such as the mouth interior or the side of the head may have simply not been seen in the available images. Next, animating the head requires plausibly morphing the reconstructed face without ever having seen that particular person talking or sticking their tongue out. Finally, the input images may be inconsistent, e.g., because the person could not remain completely static during a casual capture. All of these three challenges, i.e., sparse 3D reconstruction, face animation, and robust reconstruction, have to be solved simultaneously to successfully reconstruct a photo-realistic 3D head avatar given just a few images.

Due to the aforementioned challenges, the current best methods for high-quality avatar creation [23, 30, 36, 49, 51, 58, 59, 70, 85] cannot be applied since their optimization-based approach requires many more observations than just a few images of a person. On the other hand, methods that aim at 3D-aware portrait animation [13, 14, 20, 62] learn a prior over 3D heads from large video datasets. As a result, existing methods can already convincingly animate faces from as few as just a single input image. However, such methods typically focus on frontal facial appearance and may sacrifice 3D-consistency or temporal consistency to reach satisfying image synthesis quality. This makes these methods less applicable in scenarios where a proper 3D model is required. Another approach is to build a photo-realistic 3D head model that learns the distribution of 3D face geometry, expression, and appearance, akin to classic 3DMMs. While such unconditional generative models are very versatile and recent works have shown promising results [72, 81], they typically suffer from limited generalizability along the identity axis due to existing 3D face datasets only providing a few hundred different persons [35, 46, 48, 73].

Our approach is motivated by the observation that 3D data for faces is limited in the identity axis, but not in the expression axis that is already well captured in existing datasets. We therefore design a system that is *conditional* on the identity of a person, but *generalizes* along the expression axis. Opposed to photo-realistic 3D face models, the network does not have to learn the full extent of human facial appearance but just needs to understand how to reconstruct it from a few example images, which simplifies the task. Inspired by recent Large Reconstruction Models [56, 71, 78], we devise an architecture that predicts 3D Gaussians for each pixel in the input images. As such, we deliberately avoid anchoring the 3D Gaussians on a template mesh, as common in the 3D Head Avatar domain [6, 49, 51]. The reason is that human facial appearance is complex and di-

verse. Limiting the 3D representation to a fixed number of Gaussians anchored on a template mesh with uniform topology will cause issues in regions such as hair or face accessories. Instead, when predicting Gaussians for each foreground pixel, a person with voluminous hair will receive more primitives to model their head than a bald person. To simplify the task of sparse 3D reconstruction, we enrich each input image with feature maps from Sapiens [33] and position maps from DUST3R [65]. These position maps are injected into the network via skip connections and act as a coarse starting position for each Gaussian. We find that DUST3R is still capable of producing reasonable position maps even if the input images are not consistent. We exploit this remarkable capability of DUST3R to train our model on input images taken at different timesteps, which not only enables training and inference on larger monocular video datasets but also makes the model more robust to inconsistencies in the input images. Such inconsistencies can quickly occur in a real-life scenario where a person may not stay completely still during a phone capture of their head. Finally, to model the animation of the face, we find that simple cross-attention between intermediate feature maps and a descriptive expression code are already sufficient to generalize over the space of facial expressions. Taken together, our contributions are as follows:

- We design Avat3r, a novel pipeline for creating high-quality 3D head avatars from just a few images in a matter of minutes.
- We build the first animatable large 3D reconstruction model and show that simple cross-attention on expression codes is sufficient to model complex facial animations.
- To improve robustness, we integrate priors from Dust3r and Sapiens via skip connections, and feed input images from different timesteps to the network during training.

2. Related Work

2.1. Large 3D Reconstruction Models

Recently, generalized feed-forward models for 3D reconstruction from sparse input views have garnered considerable attention due to their applicability in heavily under-constrained scenarios. The Large Reconstruction Model (LRM) [28] uses a transformer-based encoder-decoder pipeline to infer a NeRF reconstruction from just a single image. Newer iterations have shifted the focus towards generating 3D Gaussian representations from four input images [10, 12, 42, 56, 71, 78], showing remarkable novel view synthesis results. Very recently, FaceLift [45] has proposed a combination of personalized multi-view image diffusion and a large reconstruction model for single-image 3D head reconstruction. The paradigm of transformer-based sparse 3D reconstruction has also successfully been applied to lifting monocular videos to 4D [50].

Yet, none of the existing works in the domain have studied the use-case of inferring *animatable* 3D representations from sparse input images, which is the focus of our work.

2.2. Monocular 3D Head Avatar Reconstruction

A popular task is to reconstruct an animatable 3D head avatar from a monocular input video. Several methods have been proposed based on meshes [24], NeRFs [22, 69, 86], SDFs [82], points [83], and Gaussians [11, 67]. However, due to their optimization-based nature, these approaches often exhibit poor viewpoint extrapolation capabilities due to overfitting on the input viewpoint. Very recently, Tang et al. [55] proposed to improve the geometric fidelity of reconstructed monocular avatars by incorporating priors from a personalized image diffusion model, still requiring a full monocular input video. In contrast, our method requires only 4 input images to reconstruct a complete and animatable 3D head with extensive viewpoint coverage.

2.3. 3D-aware Portrait Animation

A different line of work focuses on animating portraits in a 3D-aware manner, based on 3D volumes [20], neural rendering [60], meshes [32], TriPlanes [14, 17, 18, 61, 62], or 3D Gaussians [13]. Typically, the animation control is injected via a 3D morphable model (3DMM) [4, 40]. Concurrent to our work, CAP4D [57] uses personalized multi-view image diffusion and LAM [26] uses cross attention to build a 3D head avatar from a single image. While the aforementioned methods can animate a single portrait, they mostly focus on image synthesis from a frontal camera, often trading 3D consistency for better image quality by using 2D screen-space neural renderers. In contrast, our work aims to produce a truthful and complete 3D avatar representation from the input images that can be viewed from any angle.

2.4. Photo-realistic 3D Face Models

The increasing availability of large-scale multi-view face datasets [5, 35, 46, 48, 73] has enabled building photo-realistic 3D face models that learn a detailed prior over both geometry and appearance of human faces. HeadNeRF [27] conditions a Neural Radiance Field (NeRF) [47] on identity, expression, albedo, and illumination codes. VRMM [74] builds a high-quality and relightable 3D face model using volumetric primitives [44]. One2Avatar [76] extends a 3DMM by anchoring a radiance field to its surface. More recently, GPHM [72] and HeadGAP [81] have adopted 3D Gaussians to build a photo-realistic 3D face model. Another line of work utilizes 3D Generative Adversarial Networks [7, 8, 25] to learn the distribution of animatable head avatars [2, 38, 53, 54] from frontal 2D image datasets.

To obtain the final 3D avatar, typically a costly fitting process is necessary, impeding casual use-cases on consumer-grade devices. While this limitation may be circumvented

by learning a generalized encoder that maps images into the 3D face model’s latent space [3, 43, 52, 77, 80], another fundamental limitation remains. Even with more multi-view face datasets being published, the number of available training subjects rarely exceeds the thousands, making it hard to truly learn the full distribution of human facial appearance. 3DGAN-based approaches mitigate this by training on single image datasets, but typically struggle with view extrapolation. In this paper, we pursue a different approach that avoids generalizing over the identity axis, by conditioning on some images of a person, and only generalizes over the expression axis, for which plenty of data is available.

A similar motivation has inspired recent work on codec avatars where a generalized network infers an animatable 3D representation given a registered mesh of a person [6, 39]. The resulting avatars exhibit excellent quality at the cost of several minutes of video capture per subject and expensive test-time optimization. For example, URAvatar [39] finetunes their network on the given video recording for 3 hours on 8 A100 GPUs, making inference on consumer-grade devices impossible. In contrast, our approach directly regresses the final 3D head avatar from just four input images without the need for expensive test-time fine-tuning.

3. Method

The goal of Avat3r is to predict a high-quality 3D Gaussian representation of a human head from just a few posed input images while simultaneously animating the 3D head to match a desired facial expression:

$$\mathcal{G} = \text{AVAT3R}(I, \pi, z_{exp}) \quad (1)$$

where $I = \{I_1, \dots, I_V\}$ are the V input images, π their corresponding camera parameters, and z_{exp} a code describing the desired facial expression, such as an expression code of FLAME [40]. In our experiments, $V = 4$. Note that the images I do not necessarily have to stem from multi-view camera systems or synthetic multi-view renderings of objects as it is common for large reconstruction models. Instead, the input images may show the scene at different instances in time, making the model amenable for training and inference on frames from monocular videos.

Under the hood, Avat3r employs a Vision Transformer backbone [19] to predict 3D Gaussian primitives for each pixel in the input images. Fig. 2 depicts the model architecture. In the following, we introduce the different components of our pipeline.

3.1. Pre-trained Foundation Models

We simplify the task by employing two foundation models: DUST3R [65], and Sapiens [33]. DUST3R predicts dense

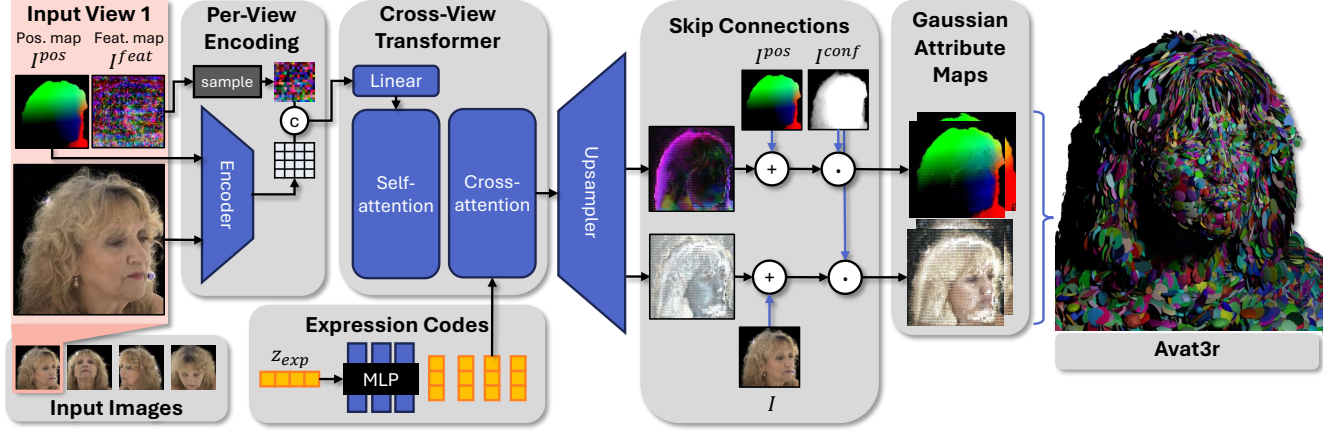


Figure 2. **Method Overview:** Avat3r reconstructs a high-quality 3D representation from just a few input images by predicting 3D Gaussian attributes for each input pixel. We first obtain position maps I^{pos} from DUST3R and feature maps I^{feat} from Sapiens for each view. These are then patchified to serve as image tokens for the Vision Transformer back-end. Dense Self-attention performs matching within tokens of the same image and across views to infer 3D structure. Dynamics are modelled via cross-attention layers that attend to a sequenced expression code. The resulting intermediate feature maps are decoded into Gaussian position, scale, rotation, color, and opacity maps, and then upsampled to the original input image resolution. Finally, we add skip connections to the predicted position and color maps. Gaussians that belong to pixels with a confidence larger than a pre-defined threshold are collected and can be rendered from arbitrary viewpoints.

position maps for a set of input images and globally aligns them with the known camera parameters π . These position maps will be beneficial for placing the 3D Gaussians. The Sapiens backbone predicts a rich low-resolution feature map for each image. The semantic nature of the extracted features will simplify the view matching task that has to be performed by the subsequent Transformer.

$$I^{pos}, I^{conf} = \text{DUST3R}(I, \pi) \quad (2)$$

$$I^{feat} = \text{SAPIENS}(I) \quad (3)$$

where I^{pos}, I^{conf} are position and confidence maps predicted by DUST3R, and I^{feat} are the Sapiens feature maps.

3.2. Animatable Large 3D Reconstruction Model

We adopt a Vision Transformer architecture following GRM [71] by first patchifying the input maps $I, I^{pos}, I^{pluck} \in \mathbb{R}^{V \times H \times W \times C}$ with a convolutional layer:

$$h = \text{CONV}([I, I^{pos}, I^{pluck}]) \quad (4)$$

where I^{pluck} are the per-ray plucker coordinates stored for each pixel and $h \in \mathbb{R}^{V \times H_p \times W_p \times D}$ will be the low-resolution intermediate feature maps for each input view. Next, we incorporate the pre-computed per-image feature maps $I^{feat} \in \mathbb{R}^{V \times H_f \times W_f \times C_f}$:

$$h^{feat} = \text{GRIDSAMPLE}(I^{feat}) \quad (5)$$

$$h \leftarrow \text{LINEAR}([h, h^{feat}]) \quad (6)$$

where the grid sampling step is necessary to transform the Sapiens features maps I^{feat} to the same $H_p \times W_p$ resolution

as the intermediate feature maps h . The subsequent linear layer then squashes the concatenated feature maps back to dimension D to avoid excessive compute increase in the following transformer layers.

The core of the Vision Transformer consists of several self-attention layers that perform cross-view matching between all $V \times H_p \times W_p$ tokens to infer 3D information from the given input images:

$$h \leftarrow \text{SELFATT}(h, h) \quad (7)$$

To make Avat3r animatable, we employ several cross-attention layers that allow each image token to attend to a sequence of expression tokens following [36, 75]:

$$f_{exp} = \text{MLP}(z_{exp}) \quad (8)$$

$$h \leftarrow \text{CROSSATT}(h, f_{exp}) \quad (9)$$

where the MLP projects a single descriptive expression code $z_{exp} \in \mathbb{R}^{C_{exp}}$ to the expression code sequence $f_{exp} \in \mathbb{R}^{S \times D}$. In our case, $S = 4$. Surprisingly, by simply enriching each image token in h with expression-dependent information, Avat3r is already capable of modeling complex facial animations in 3D.

We upsample the image tokens again to the original input resolution using the transformer-based upsampler from GRM [71]:

$$M = \text{UPSAMPLE}(h) \quad (10)$$

where $M \in \mathbb{R}^{V \times H \times W \times C_G}$ denote the Gaussian attribute maps consisting of per-pixel Gaussian positions, scales, rotations, colors, and opacities. After upsampling, we add two

skip connections for positions and colors:

$$M^{pos} \leftarrow M^{pos} + I^{pos} \quad (11)$$

$$M^{rgb} \leftarrow M^{rgb} + I \quad (12)$$

which incorporate the inductive bias that a pixel should primarily spawn a 3D Gaussian that models the pixel’s corresponding 3D geometry.

Finally, we make use of DUST3R’s per-pixel confidence maps I^{conf} to decide which pixel in the generated Gaussian attribute maps M should actually spawn a 3D Gaussian:

$$\mathcal{G} = \{M[x, y] : I^{conf}[x, y] > \tau\} \quad (13)$$

where $\tau = 0.5$ is a threshold describing the minimal confidence needed to create a 3D Gaussian. This serves two purposes: First, it avoids artifacts that arise from incorrect 3D position initializations from DUST3R. Second, it naturally adjusts the number of Gaussians to the person at hand. In that sense, the confidence mask acts much like a foreground segmentation: If the person had large voluminous hair, then more pixels will belong to the foreground and thus more Gaussians will be kept.

The final set of 3D Gaussians $\mathcal{G}$ can then be rendered from any desired novel viewpoint π^{nv} using the tile-based differentiable rasterizer $\mathcal{R}$ [31]:

$$I^{nv} = \mathcal{R}(\mathcal{G}, \pi^{nv}) \quad (14)$$

3.3. Losses

Avat3r can be trained with only photometric losses on novel viewpoints following the original 3D Gaussian Splatting work [31]:

$$\mathcal{L}_{l1} = \|I^{nv} - I^{gt}\|_1 \quad (15)$$

$$\mathcal{L}_{ssim} = \text{SSIM}(I^{nv}, I^{gt}) \quad (16)$$

Furthermore, we incorporate perceptual losses [79] to encourage the emergence of more high-frequency details:

$$\mathcal{L}_{lpips} = \text{LPIPS}(I^{nv}, I^{gt}) \quad (17)$$

The final loss term is comprised as follows:

$$\mathcal{L} = \lambda_{l1}\mathcal{L}_{l1} + \lambda_{ssim}\mathcal{L}_{ssim} + \lambda_{lpips}\mathcal{L}_{lpips} \quad (18)$$

with $\lambda_{l1} = 0.8$, $\lambda_{ssim} = 0.2$, and $\lambda_{lpips} = 0.01$.

4. Experimental Results

4.1. Training

We train Avat3r on multi-view video captures from the Ava256 dataset [46]. In particular, we use the 4TB version of the dataset which contains 7.5fps videos from 80 cameras (roughly 5000 frames each) at a resolution of 1024×667 .

Avat3r is trained on 244 persons from the dataset with the remaining 12 being held out. We use the provided tracked mesh to crop each image to a head-centric 512×512 portrait which will be fed as input to our model. For each training sample, we randomly select 4 reasonably diverse viewpoints and 4 timesteps with diverse expressions as input images I . On the other hand, the supervision images I^{gt} are left at the highest resolution and are randomly sampled over all available cameras and timesteps. For z_{exp} , we use the expression codes that are provided with the Ava256 dataset.

Avat3r is trained with Adam [34] using a learning rate of $5e-5$. We only introduce $\mathcal{L}_{lpips}$ after 3M optimization steps to avoid focusing on high-frequency details too early. In total, the model is trained for 3.5M steps with a batch size of 1 per GPU on 8 A100 GPUs which takes roughly 4 days. We further find that training convergence is significantly improved when supervising multiple viewpoints for each batch. I.e., a batch consists of 12 images: 4 with random expression that are used as input, and 8 viewpoints from the target expression that are used as supervision.

4.2. Experiment Setup

Task. We evaluate the ability to create a 3D head avatar from four images of an unseen person.

Metrics. We employ three paired-image metrics to measure the quality of individual rendered images: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) [66], and Learned Perceptual Image Patch Similarity (LPIPS) [79]. Furthermore, we make use of two face-specific metrics: Average Keypoint Distance (AKD) measured in pixels with keypoints estimated from PIPNet [29], and cosine similarity (CSIM) of identity embeddings based on ArcFace [16].

Baselines. We compare our method with recent state-of-the-art systems for 3D head avatar creation from few input images.

HeadNeRF [27]. A NeRF-based Autodecoder with disentangled identity and expression spaces that can be fitted to an arbitrary number of images.

InvertAvatar [80]. A feed-forward encoder that predicts the underlying representation of Next3D [53] from 4 input images which then can be animated.

GPAvatar [14]. A NeRF-based method that predicts canonical TriPlanes from multiple input images. Animations are then modeled by querying 3D points in an expression-dependent feature field inferred from FLAME.

4.3. Few-shot 3D Head Avatar Creation

We compare the ability to create a realistic 3D head avatar from just four input images on both the Ava256 dataset [46]



Figure 3. **Comparison for few-shot 3D Head Avatar creation.** We evaluate the ability of Avat3r to create 3D head avatars from four input images in a self-reenactment setting. Note that the NeRSemble dataset has not been used during training of Avat3r and therefore both source and driver person are out-of-domain in this case. Nevertheless, our method produces high quality avatars.

and the NeRSemble dataset [35]. To this end, we train GPAvatar on the Ava256 dataset in the same setting as our model and use the high-quality tracked meshes provided with the dataset to drive GPAvatar. For HeadNeRF and InvertAvatar, we use the published code bases for reconstruction, animation and rendering. We evaluate all methods at 512×512 pixels resolution on the same hold-out persons. For evaluation on Ava256, we sample 1000 different target expressions from 12 persons from different viewpoints. On NeRSemble, we sample 100 target images from 3 male and 3 female subjects.

The quantitative results are listed in Tab. 1. It can be seen that our method clearly outperforms the baselines. We observe large differences in metrics measuring alignment (PSNR, SSIM), identity similarity (CSIM), and facial key-point similarity (AKD). This indicates that Avat3r produces 3D head avatars that better resemble the person in the input images and produces more realistic animations. Note that HeadNeRF cannot render any other background than white. We therefore compute metrics against ground truth images with white background for that method. We further

show qualitative comparisons in Fig. 3. Our method produces more expressive avatars with higher rendering quality than the baselines and even performs well on a person with colored hair who wears a headband. Also, note that the NeRSemble dataset has not been used during training and the images in this dataset exhibit entirely different characteristics in terms of lighting, camera parameters, and viewpoints, compared to the training images from the Ava256 dataset. Nevertheless, Avat3r generates realistic 3D head avatars in this setting as well.

For video renderings and single-image results, see our supplemental material.

4.4. Ablations

In the following, we study the efficacy of our design choices for Avat3r. In particular, we look at the usefulness of DUST3R’s position maps, the Sapiens feature maps, and using input images with different expressions during training. More ablations and a runtime analysis can be found in the supplemental material. Our findings are summarized in Tab. 2 and Fig. 4.

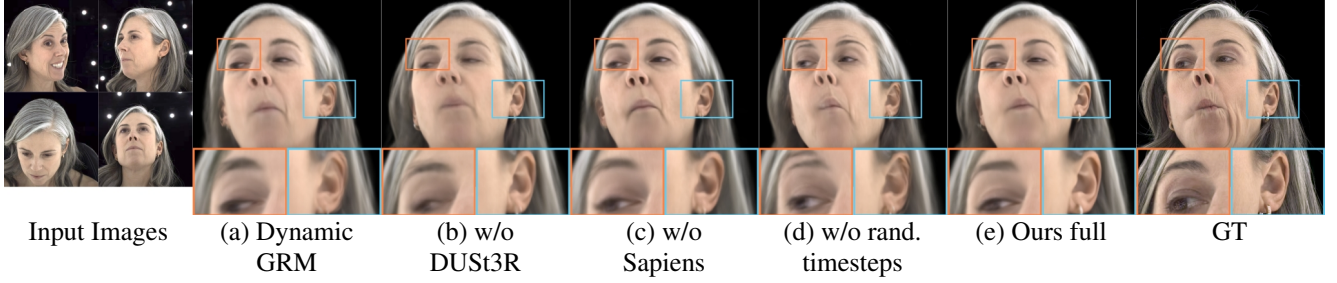


Figure 4. **Ablation study.** Adding Sapiens features (b) noticeably improves sharpness but still exhibits misalignments. Conversely, using position maps from Dust3r improves alignment but is overall less sharp (c). Combining both but training on multi-view images from the same timestep produces sharp results but is less robust to inconsistencies in the input (d). Our final model exhibits the best trade-off between alignment, sharpness, and robustness (e).

		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	AKD $\downarrow$	CSIM $\uparrow$
Ava256	HeadNeRF	9.1	0.64	0.52	6.9	0.11
	InvertAvatar	14.2	0.36	0.55	15.8	0.29
	GPAvatar	19.4	0.69	0.34	5.3	0.31
	Ours	20.7	0.71	0.33	4.8	0.59
NeRSemble	HeadNeRF	9.8	0.69	0.47	4.9	0.22
	InvertAvatar	17.2	0.44	0.49	5.6	0.44
	GPAvatar †	17.6	0.67	0.40	5.7	0.07
	Ours	20.5	0.75	0.33	3.7	0.50

Table 1. **3D Head Avatar Creation Comparison.** Our method outperforms the baselines in all metrics.

	$\mathcal{S}$	$\mathcal{D}$	$\mathcal{T}$	PSNR $\uparrow$	LPIPS $\downarrow$	AKD $\downarrow$
Dynamic GRM	$\square$	$\square$	$\boxtimes$	20.8	0.439	8.60
w/o Sapiens	$\square$	$\boxtimes$	$\boxtimes$	20.9	0.434	8.59
w/o Dust3r	$\boxtimes$	$\square$	$\boxtimes$	21.1	0.429	9.60
w/o rand. timesteps	$\boxtimes$	$\boxtimes$	$\square$	21.3	0.409	8.86
Avat3r	$\boxtimes$	$\boxtimes$	$\boxtimes$	21.6	0.410	8.08

Table 2. **Ablation.** We analyze the effect of Sapiens features ($\mathcal{S}$), DUST3R position maps ($\mathcal{D}$), and training on randomized input timesteps ($\mathcal{T}$). All ablation models are trained without LPIPS loss. Metrics are computed on 667×667 renderings.

Effect of DUST3R position maps. Removing DUST3R from our pipeline mostly impairs geometric fidelity. In Fig. 4, it can be seen that Avat3r without DUST3R struggles with aligning the Gaussian predictions from the four input images.

Effect of Sapiens feature maps. Sapiens provides rich semantic information for the input images that mostly help in producing sharper predictions, e.g., in the hair area.

Effect of training with inconsistent input images. A model that is trained on only multi-view consistent input images produces slightly sharper images. However, any inconsistencies in the input images, which can easily happen in practice, carry over to the generated 3D representation causing unwanted artifacts.

4.5. Additional Results

Extreme expressions. We showcase the generalization capability of our method by interpolating between two extreme expressions in Fig. 5. Both source and driver person are unseen.

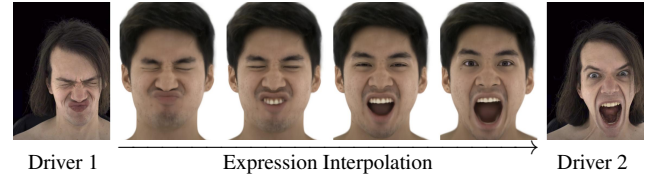


Figure 5. **Expression Generalization.**

Adapting to arbitrary input views. Despite only being trained with 4 input views, our model can be directly used when more or fewer views are available, see Fig. 6.

4.6. Applications

To make Avat3r amenable for inference on only a single input image, we make use of a pre-trained 3D GAN [77] to first lift the single image to 3D and then render four views of the head. In Fig. 7, we show three use-cases of Avat3r: (i) Generating a novel 3D head with GGHead [37] and then animating it with Avat3r, (ii) Generating a portrait image with a text-to-image diffusion model [15], lifting it with a 3D GAN [77], and animating it with Avat3r, and (iii) Lifting an existing photograph with a 3D GAN [77] and animating it with Avat3r. The results demonstrate that, despite being designed for a few-shot scenario and only being trained on real studio data, Avat3r can be used in a variety of ca-

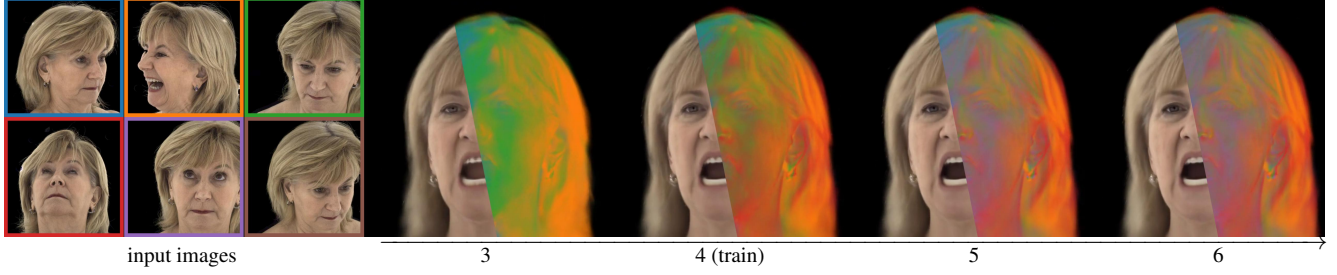


Figure 6. **Input Flexibility.** Our model accepts different numbers of views. Colors indicate from which view a Gaussian originates.

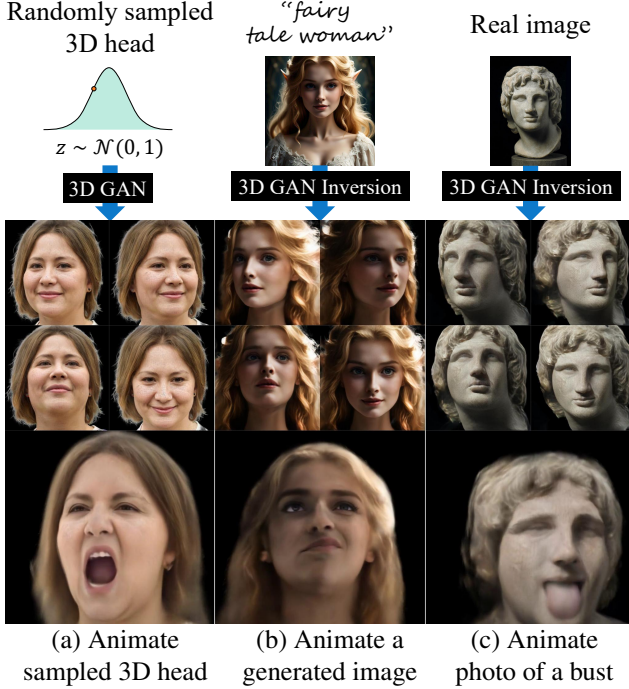


Figure 7. **Application scenarios of Avat3r.** We demonstrate various use-cases of our model such as animating a randomly sampled 3D head, animating an image generated by a text-to-image diffusion model, and animating a picture of a greek bust.

sual situations. For example, our findings for case (i) suggest that Avat3r may be used to animate images from any 3D GAN that generates 3D heads, effectively making any purely static 3D generation model animatable.

4.7. Limitations

Despite Avat3r’s great ability to create 3D head avatars from a few or even just a single input image, several limitations remain. For example, the current pipeline for inferring avatars from a single image requires a 3D GAN for 3D lifting which accumulates errors in the process in two ways: First, the 3D GAN cannot perfectly explain the person, inevitably losing some details, and second, existing inversion frameworks employ NeRF-based 3D GANs that rely on screen-space super-resolution after rendering which

introduces view-inconsistencies to the final images. Both aspects hamper the quality of the final avatar. However, these issues will decrease as newer and better 3D GANs become available that do not require screen-space rendering anymore [37, 63]. Another limitation of our framework is the requirement of camera poses during inference. While reasonable estimates for face images can be obtained by head tracking algorithms [84], incorrect camera estimates still remain a potential source of error for real use-cases. Possible remedies may include teaching the network to deal with incorrect camera estimates already during training, or dropping the requirement for camera poses altogether [64]. Finally, our current pipeline does not provide control over the lighting and instead bakes in light effects from the input images. This limits applications that put avatars into novel environments since the generated head will look out of place. Addressing this issue by explicitly disentangling the lighting properties is an exciting avenue for future research.

5. Conclusion

We have presented Avat3r, a system for directly regressing an animatable 3D head avatar from just four input images. Our pipeline combines recent advancements in large reconstruction models with the powerful foundation models DUST3R and Sapiens to obtain high quality results, and models dynamics with simple cross attention layers. Moreover, we demonstrate that Avat3r can also be employed in a single image scenario by using a pre-trained 3D GAN. Our experiments show that Avat3r learns a powerful prior over 3D facial movement that even generalizes to out-of-distribution examples like pictures of antique busts or AI-generated images. We believe that our findings open up interesting avenues for future research. For example, we have shown that LRM-like architectures have the ability to infer 3D representations even from inconsistent input images such as frames of a monocular video. This opens up the possibility to train models on large-scale monocular video datasets with a much larger variety of identities. Finally, the underlying architecture could also be used as a 3D lifting and denoising module in a diffusion framework [1, 9], enabling training of diffusion models on monocular video data for unconditional 3D head avatar generation.

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Avat3r: Large Animatable Gaussian Reconstruction Model for High-fidelity 3D Head Avatars

Supplementary Material

Inputs		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	AKD $\downarrow$	CSIM $\uparrow$
Ava256	1 GAGAvatar	18.1	0.66	0.37	7.0	0.45
	1 GAGAvatar [†]	19.6	0.68	0.31	5.6	0.33
	1 Ours ^{3DGAN}	19.1	0.68	0.39	6.0	0.37
	1 Ours ¹	19.6	0.70	0.46	5.7	0.40
Video	FlashAvatar	15.0	0.42	0.61	8.8	0.17
	4 Ours	20.5	0.75	0.33	3.7	0.50
	4 Ours ⁺⁹⁸⁴	21.6	0.77	0.29	3.8	0.76
NeRSemble	1 HeadNeRF	9.7	0.69	0.48	5.2	0.18
	1 Portrait4D-v2	17.5	0.58	0.36	5.4	0.41
	1 GAGAvatar	18.7	0.70	0.35	5.4	0.44
	1 GAGAvatar [†]	18.9	0.70	0.32	4.8	0.24
	1 Ours ^{3DGAN}	19.5	0.71	0.38	4.3	0.31
	1 Ours ^{3DGAN + 984}	19.4	0.71	0.38	4.6	0.46
	1 Ours ¹	19.8	0.73	0.38	4.4	0.30

[†]re-trained on Ava256

¹trained on 1 input view

Table 3. **Quantitative Comparison on 3D head avatar creation from various.** Our approach performs competitively when only a single input image is available, despite not being designed for a single image use-case. It also performs much better compared to monocular methods that receive a full video as input. Ours^{3DGAN} denotes a 4-shot model where the 4 required images are obtained from the single input via 3D lifting with a 3D GAN. Ours⁺⁹⁸⁴ denotes a model that is additionally fine-tuned on 984 neutral identities for better identity preservation (CSIM metric).

A. Additional Results

A.1. Avat3rs from Phone Captures and Accessories

Fig. 12 showcases additional phone scans captured by users on their own devices, including challenging cases with glasses, rotated inputs, and a NeRSemble subject wearing a headscarf. Despite not being trained on accessories, the model handles glasses reasonably well and reconstructs the headscarf with high fidelity.

A.2. Single-shot 3D Head Avatar Creation

To make Avat3r amenable for inference on only a single input image, we make use of a pre-trained 3D GAN [77] to first lift the single image to 3D and then render four views of the head. These renderings then constitute the input for Avat3r. We conduct comparisons with the recent 3D-aware portrait animation method GAGAvatar [13]. Specifically, we compare with two version of GAGAvatar: One provided by the authors which is trained on VFHQ [68], and

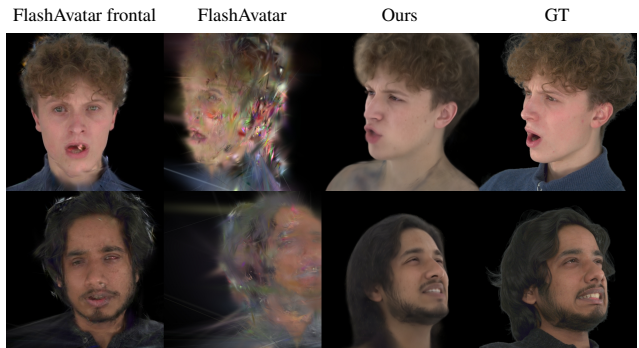


Figure 9. Comparison with FlashAvatar on NeRSemble.

another version, denoted as GAGAvatar[†], that we train on the Ava256 dataset in the same setting as our method. To drive GAGAvatar, we use their monocular FLAME tracker to obtain tracked meshes. We also compare with another 3D portrait animation method, Portrait4Dv2 [18], and HeadNeRF [27]. Fig. 10 and Fig. 11 show qualitative comparisons between our method and the baselines for single input images of hold out persons. Note that our method performs competitively compared to the single-input baselines despite never being trained for a single-shot scenario. We also include a version of our model that was trained on a single input image without DUST3R. In general, we find that Avat3r produces more realistic facial expressions than GAGAvatar and Portrait4Dv2 which are limited by FLAME’s expression space. Furthermore, our method allows much more extreme viewpoint changes without sacrificing rendering quality. Tab. 3 shows quantitative results. Note that, in contrast to portrait animation methods like GAGAvatar and Portrait4Dv2, our method can benefit when more input views are available (see Sec. B and tab. 1 in the main paper).

A.3. Comparison with Monocular Methods

We compare with the recent monocular approach FlashAvatar and provide it with a full video sequence from NeRSemble. As shown in Fig. 9, the monocular method performs well from the training view but fails on novel views due to overfitting. Please also refer to the Gaussian Avatar Fusion or HeadGAP papers, which made similar observations. In contrast, our learned reconstruction prior produces a plausible 3D head avatar. Quantitative comparisons are shown in Tab. 3.

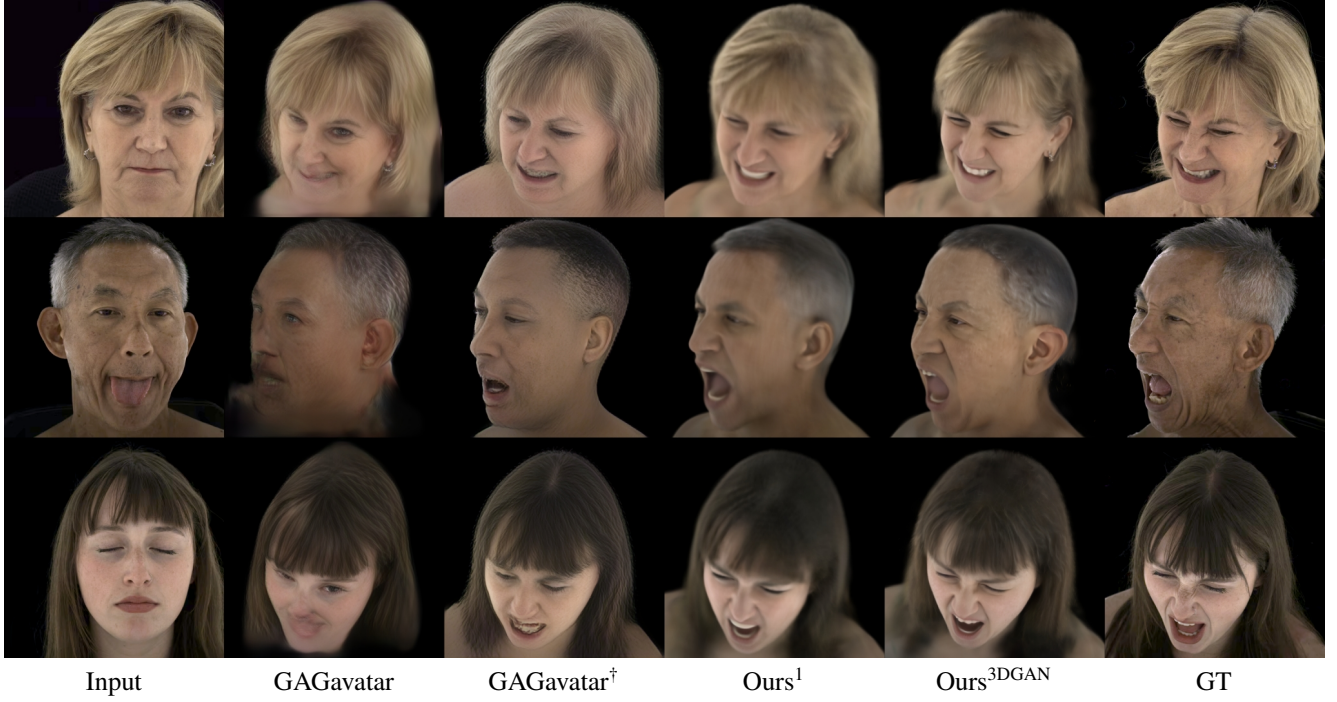


Figure 10. **Single-image comparison on Ava256.** We compare Avat3r with the recent 3D-aware portrait animation method GAGAvatar [13] in a self-reenactment scenario on hold-out persons from the Ava256 dataset. GAGAvatar[†] denotes a version of the baseline that we trained on the Ava256 dataset. Ours¹ is a version of our model that was trained on only 1 input image (see Sec. B). Our method with 3D lifting (Ours^{3DGAN}) shows better rendering quality than the baselines, especially for extreme expressions and viewing angles.

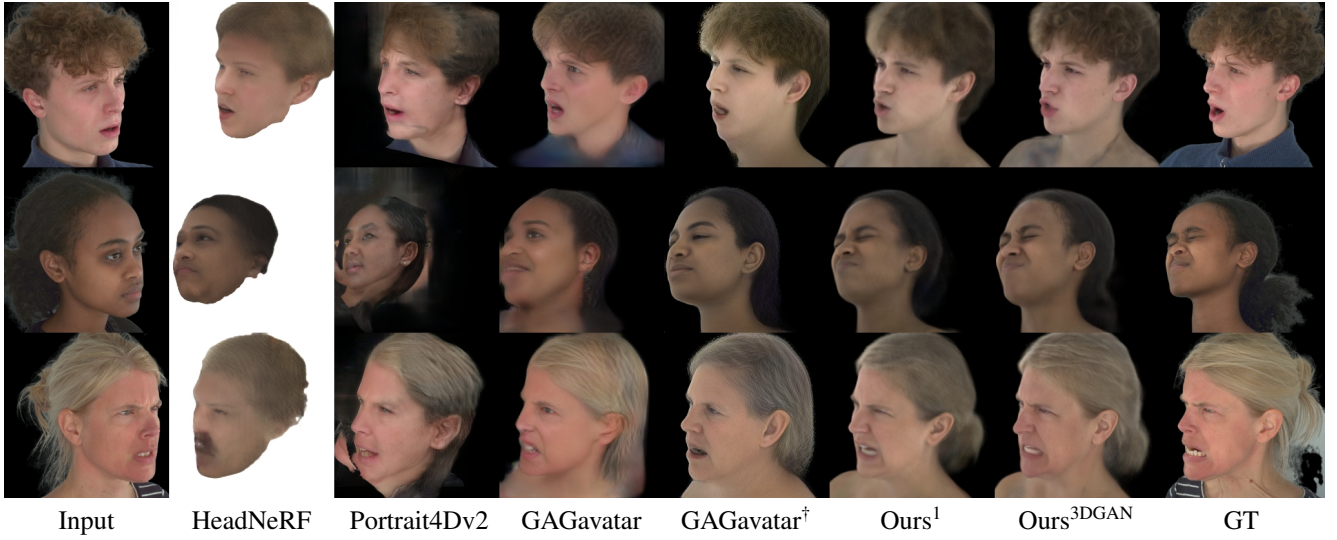


Figure 11. **Single-image comparison on NeRSemble.** We compare Avat3r with the recent 3D-aware portrait animation methods GAGAvatar [13] (GAGAvatar[†] denotes a version of the baseline that we trained on the Ava256 dataset) and Portrait4Dv2 [18] as well as the NeRF-based face model HeadNeRF [27] in a self-reenactment scenario on persons from the NeRSemble dataset [35]. Note that the NeRSemble dataset has not been used during training and therefore constitutes an evaluation scenario where both source and driver image are out-of-domain. Ours¹ is a version of our model that was trained on only 1 input image (see Sec. B). Our method with 3D lifting (Ours^{3DGAN}) shows better rendering quality than the baselines in these challenging scenarios where input and target view are from opposite sides of the face.



Figure 12. **More Avat3r reconstruction results** from phone scans with different phone cameras and persons with challenging clothing.

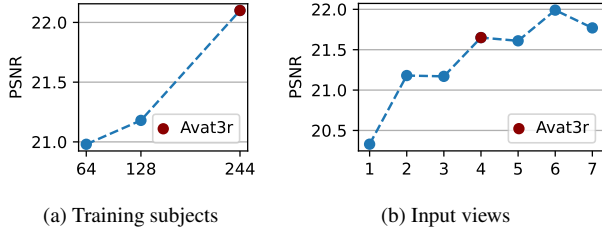


Figure 13. **Analysis of Data Efficiency.** We study how reconstruction and animation performance behaves when changing the number of training subjects and input views.



Figure 14. **Effect of Number of Train Subjects.** Training on a larger and more diverse set of people enhances the Avat3r’s generalization capabilities, as expected. This leads to more accurate reconstructions, with avatars better matching the identities shown in the input images. For instance, when dealing with complex hairstyles, a model trained on a broader range of individuals reproduces the hairstyle more accurately. All ablations are trained without LPIPS loss.



Figure 15. **Effect of Number of Input Views.** Training with just a single input image noticeably impairs quality. On the other hand, using more than 4 input images during training does not lead to significant improvements. Models are trained without DUST3R position maps and without LPIPS loss in the interest of comparability.



Figure 16. **Improved identity preservation** by including 984 additional identities with neutral expression during training.

B. Analysis of Data Efficiency

B.1. Scaling Subjects and Views from Ava256

In Fig. 13 we show how our model scales with the number of training subjects and input views available in the Ava256 dataset. For the analysis on the number of input views, we disable DUST3R as it produces less reliable position maps for 2 input views, and cannot be executed at all for 1 input view. We see a clear improvement when using more training subjects as well as using more input views. However, further scaling the number of input views also has drawbacks, as it drastically increases runtime due to dense attention inside the transformer and the increased number of Gaussians that have to be rendered. We qualitatively analyze the effect of using more train subjects in Fig. 14 and the effect of the number of input views in Fig. 15.

B.2. Effect of Adding More Neutral Subjects

Since Avat3r is only trained on 244 subjects from the Ava256 dataset, it is at risk of overfitting to those identities during training time. In our experiments, we observe that while there is a slight identity shift between the final avatar and the person in the input images, the expression transfer works quite well. We therefore hypothesize that for further improvements on the model’s generalization capabilities it is not necessary to also add thousands of expressions for each additional person. To test this, we fine-tune our model with 984 additional identities from an internal dataset — just one expression each — adding only 0.08% to the training data. As shown in Fig. 16 and confirmed by improved CSIM metrics in Tab. 3, this small addition noticeably improves identity retention for these challenging cases.



Figure 17. **Qualitative Ablation of Skip Connections.** Not employing skip connections (a) causes misalignments, blurry renderings, and a slight color shift. Adding the color skip connection (b) already noticeably improves sharpness and color fidelity. On the other hand, if the position skip connection is added (c), geometric details are improved but the overall color slightly off. Using both skip connections (d) yields the best result.

	Creation↓ in [s]	Driving↑ in [fps]
HeadNeRF	65	111
Portrait4D-v2	0.2	4
GAGAvatar	0.1	63
GPAvatar	0.2	9.5
Ours ¹	1.1	53
Ours ^{3DGAN}	17.9	7.9
Ours	12.3	7.9

Table 4. **Runtime analysis.** Our method can create a high-quality avatar in a few seconds, and animate it at interactive rates.

	$\mathcal{C}$	$\mathcal{P}$	PSNR↑	SSIM↑	LPIPS↓	JOD↑	AKD↓	CSIM↑
No skip	☐	☐	21.39	0.740	0.456	4.99	9.24	0.60
No pos. skip	☒	☐	21.76	0.746	0.443	5.03	9.04	0.611
No col. skip	☐	☒	21.55	0.745	0.435	5.00	7.69	0.648
Avat3r	☒	☒	22.05	0.751	0.421	5.15	7.99	0.689

Table 5. **Quantitative Ablation of Skip Connections.** We analyze the effect of the color ($\mathcal{C}$) and position ($\mathcal{P}$) skip connections. All ablation models are trained without LPIPS loss. Metrics are computed on 667×667 renderings.

C. Inference efficiency & driving speed.

While Avatar creation takes several seconds for our method due to DuSt3R and Sapiens, we can cache all activations up to the final cross-attention layers afterwards, leading to expression driving at 7.9fps for our 4-shot model and 53fps for our single-shot model, see Tab. 4. Runtimes measured on a single RTX3090 GPU.

D. Effect of Skip Connections

We analyze the effect of the proposed skip connections, i.e., omitting Eq. (11), Eq. (12), or both of the main paper. The results are listed in Tab. 5. We observe a noticeable hit in performance when either skip connection is removed. Furthermore, we qualitatively analyze the effect of skip connections in Fig. 17.

E. Training Details

Dataset processing. We use the 4TB version of the Ava256 dataset [46] which contains 256 persons, 80 cameras, and roughly 5000 frames per person that are sampled at 7.5 fps. We compute foreground segmentation masks with BackgroundMattingV2 [41] and replace the background in all images with black pixels. We use the provided tracked mesh to find a 512×512 head-centered square crop for input images and 667×667 head-centered square crop for supervision views. This ensures that the pixels in the input images are used efficiently to show as much as possible of the head, leading to more 3D Gaussians. The reason for also cropping the target images is to remove parts of the torso, as it is not the focus of this work.

DUST3R and Sapiens. Since both DUST3R [65] and Sapiens [33] are expensive foundation models, we pre-compute the position and feature maps for the input frames. For Dust3r, it is prohibitive to pre-compute all possible combinations of 4 input views out of the available 80 cameras. Instead, we choose 3 "reasonable partner views" for each input and only store the position map for that viewpoint. This assigns each input view exactly one position map, which is conceptually wrong since the position map from DUST3R should depend on the other 3 selected views. Nevertheless, we did not observe any disadvantages from this simplification strategy.

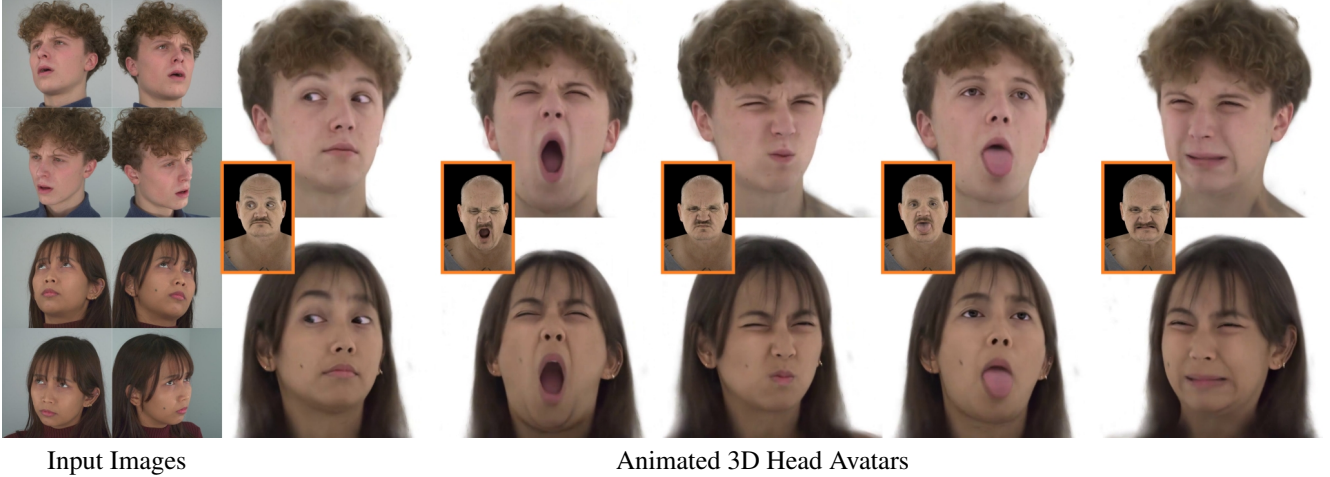


Figure 18. **Performance on NeRSemble dataset.** We show reconstructed and animated avatars using 4 images from the NeRSemble dataset. Note that this dataset was not used during training and contains images with different lighting conditions, viewpoints, and camera intrinsics than the Ava-256 dataset that was used to train Avat3r.

	Hyperparameter	Value
ViT & GRM	ViT patch size	8×8
	hidden dimension D	768
	#self-attention layers	8
	#cross-attention layers	8
	#GRM transformer upsampler step	1
Input & Output	Sapiens version	2b
	Sapiens feature dimension	1920
	Input image resolution	512×512
	Gaussian attribute map resolution	512×512
	Train render resolution	667×667
Expression MLP	Dimension of expression code	256
	#expression sequence MLP layers	2
	Dimension of expression sequence MLP	256
	Expression sequence MLP activation	ReLU

Table 6. **Hyperparameters.**

Head-centric coordinates. We further simplify the task by factoring the head poses from the provided tracked mesh into the camera poses instead of letting the network predict them. That way, our model can always predict the head in canonical pose, making the task easier. This is possible because modeling the torso, which in head-centric coordinates moves a lot when the person shakes their head, is not the focus of this work.

Expression codes. Our architecture is agnostic to the specific choice of animation signal. For experiments on ava256, we used the dataset’s expression codes that were originally predicted by a generalized expression encoder providing a driving signal beyond FLAME’s topology. For experiments on NeRSemble and in-the-wild driving videos, we fine-tuned our model using FLAME codes obtained by running GAGAvatar’s version of Metrical Tracker [13, 84]. This shows that Avat3r learns a general notion of facial ex-

pressions that can be adapted to fit a specific driving signal.

k-farthest viewpoint sampling. To ensure that the 4 input images always follow a reasonable viewpoint distribution, we employ k-farthest viewpoint sampling. Specifically, we first start from a random camera and collect a set of 10 candidate cameras that are evenly spread out using farthest point sampling. From this candidate set, we then randomly select 4 cameras as input. This two-stage approach ensures that the input cameras are sufficiently random during training but also reasonably spread out to avoid seeing a person only from one side. During sampling input viewpoints, we exclude cameras that only observe the person from the back since those are not realistic inputs during test-time.

Input timestep sampling. To improve robustness of our model, we sample different timesteps for each of the 4 input images. This ensures that the model can deal with inconsistencies in the input. To maximize the diversity in the input expressions, we uniformly sample 10 timesteps in the segments: EXP_eye_wide, EXP_tongue001, and EXP_jaw003 from the recordings of the Ava256 dataset. This covers the most extreme facial expressions while avoiding having to pre-compute DUST3R and Sapiens maps for every single image in the dataset.

Speed. To speed-up training, we employ the 3D Gaussian Splatting performance improvements of DISTWAR [21].

F. Hyperparameters

In Tab. 6, we list the most important hyperparameters for training Avat3r.